

given by Galileo.

↑ Laws of Motion

Inertia: It is defined as the inability of a body to change its state of rest, state of motion or direction of motion by itself (on its own).

OR

It is defined as the property of a body by virtue of which it is unable to change its state of rest, state of motion or direction of motion on its own.

It is of three types

- 1 Inertia of rest: It is the inability of a body to change its state of rest on its own.

Example: A passenger in a bus is pushed backwards when the bus suddenly starts moving in forward direction.

- 2 Inertia of motion: It is the inability of a body to change its state of motion on its own.

Example: When a ~~car~~ driver in a moving bus suddenly applies ~~brakes~~ brakes. The passengers are pushed in forward direction.. due to inertia of motion.

3. Inertia of direction. It is the inability of a body to change its direction of motion on its own.

Example: When a bus moving on a straight road suddenly moves along a curve

the passengers are pushed outwards due to inertia of direction.

Newton's First law of motion.

A body at rest continues to remain at rest, a body in motion continues to remain in motion and a body moving in some direction continues to move in the same direction unless it is acted upon by some external force.

Newton's first law of motion defines inertia that is why, it is also called as law of inertia.

Newton's First law of motion defines force qualitatively.

Linear Momentum [P].

It is defined as the quantity of motion contained in a body.

If 'm' is the mass and 'v' is the velocity of a body, then the linear momentum of the body is given by

$$P = m \cdot v.$$

SI unit of $P = \text{kg ms}^{-1}$.

cgs " " $P = \text{gm s}^{-1}$.

DF of $P = M^1 L^1 T^{-1}$.

It is a vector quantity.

In vector form $\vec{P} = m\vec{v}$.

Newton's second law of motion.
It states that the force (F) applied on a body is directly proportional to the rate of change of its linear momentum.

$$\vec{F} \propto \frac{d\vec{p}}{dt}$$

$$\vec{F} = k \frac{d\vec{p}}{dt}$$

$$k = 1$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

The force applied on a body, is equal to $\vec{F} = m \times$ rate of change of its linear momentum.

$$\text{Now, } \vec{p} = m \cdot \vec{v}$$

$$\vec{F} = \frac{d(m \cdot \vec{v})}{dt}$$

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

$$\vec{F} = m \cdot a$$

The force applied on a body is equal to the product of its mass and acceleration.

K Math. w. Ford.

$$\vec{F} = m \cdot \vec{a}$$

On scalar form.

$$F = m \cdot a$$

*. If $a = 0$, velocity is constant (uniform motion)

$$F = ?$$

$$F = m(0)$$

$$F = 0$$

If the body is at rest.

$$a = 0$$

$$F = 0$$

⇒ From the equation $F = ma$, it is clear that in order to produce acceleration in a body (changing state of rest, state of motion or direction of motion), force is required which is the statement of Newton's first law also. Hence first law is contained in second law.

Newton's second law defines force.

Force can be defined as ~~an~~ an external agent which when applied on a body produces acceleration in it.

K Math. w. Ford.

1. If there is no force, there is no motion.

2. If there is no motion, there is no force.

Statement I is wrong as in uniform motion, force on the body is zero.
Statement II is correct.

$$F = ma.$$

Absolute unit

SI unit of force = kgms^{-2} = newton.

cgs unit of force = gcm^{-2} = dyne.

$$1 \text{ newton} = \text{kgms}^{-2}.$$

$$1 \text{ newton} = 1000 \text{g} \cdot 100 \text{cm} \text{s}^{-2}.$$

$$1 \text{ newton} = 10^5 \text{ dyne}.$$

$$1 \text{ newton} = 10^5 \text{ dyne}.$$

Gravitational units

SI unit of force = kgwt [wt = weight].
= kgf .

$$1 \text{ kgwt} = 1 \text{ kgf} = 9.8 \text{ N}.$$

cgs unit of force = gwt = gf .

$$1 \text{ gwt} = 1 \text{ gf} = 980 \text{ dyne}.$$

NEET
2028!

conservation of linear momentum.

we know that,

$$\vec{F} = \frac{d\vec{p}}{dt}$$

if the net force acting on the body is zero.

$$\vec{F} = 0$$

$$\Rightarrow \frac{d\vec{p}}{dt} = 0$$

$$\vec{p} = \text{constant}$$

if the net force acting on a body is zero, the linear momentum of the body remains constant. This is known as "law of conservation of linear momentum".
[Another statement]

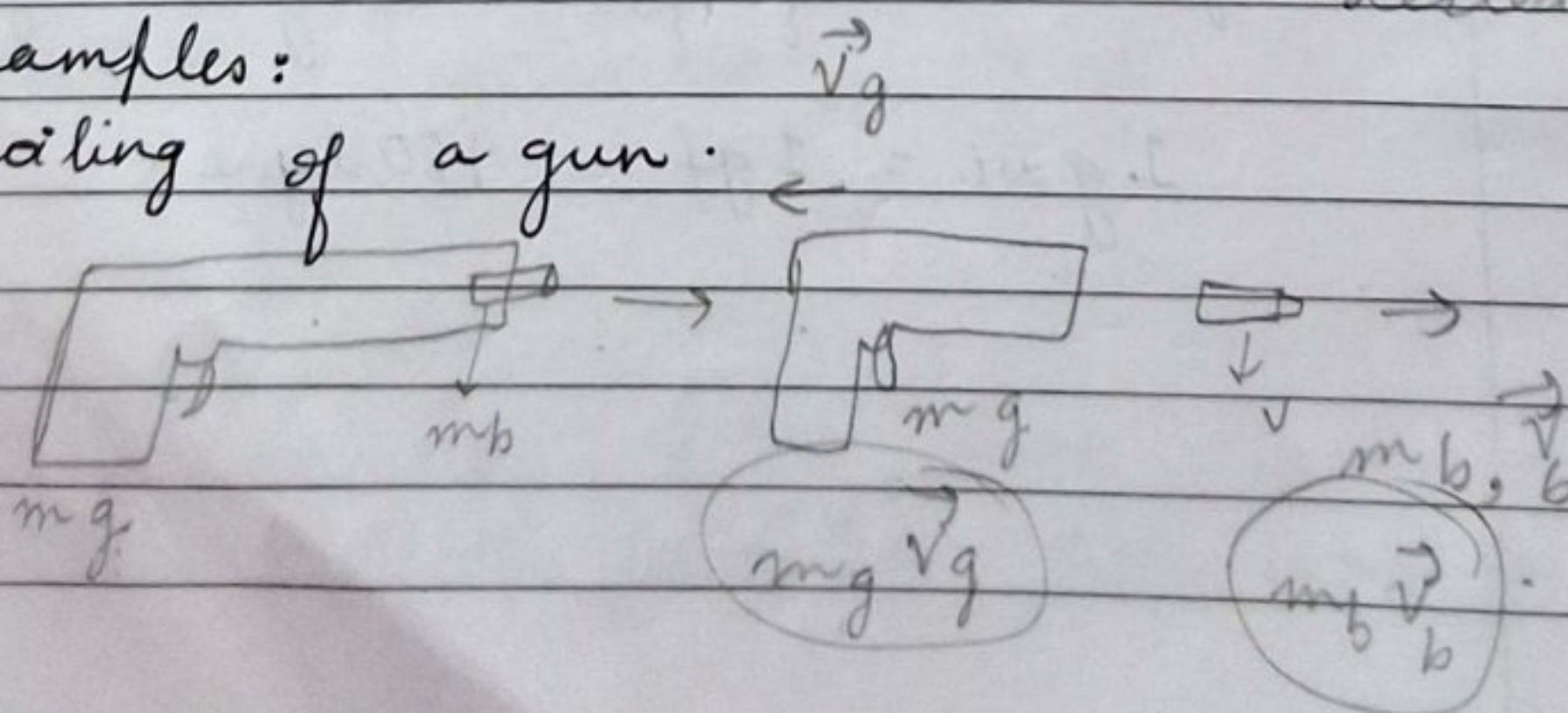
The linear momentum of an isolated system is always constant.

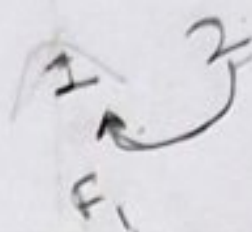
no external force is exerted.

Examples:

1

Recoiling of a gun.





linear momentum = 0.
we have.

$$0 = m_g \vec{v}_g + m_b \vec{v}_b$$

$$m_g \vec{v}_g = -m_b \vec{v}_b$$

$$\vec{v}_g = \frac{-m_b \vec{v}_b}{m_g} \rightarrow \text{velocity of recoil of gun.}$$

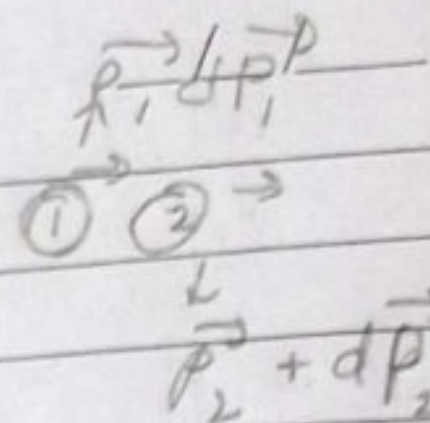
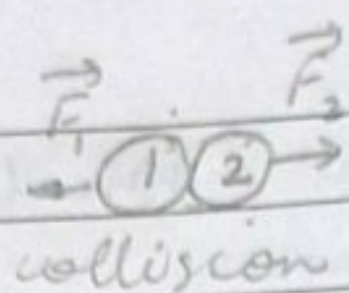
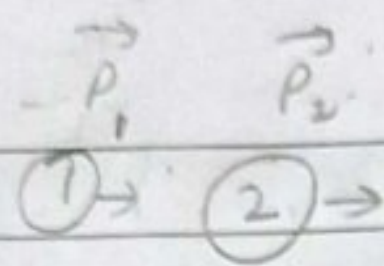
2. When one jumps out of a boat to the shore, the boat moves backwards to conserve linear momentum.

3. Propulsion of rockets works on the law of conservation of linear momentum.

3. Newton's third law of motion. It states that force never exists singly. Forces always exist in pairs and to every action (force) there is an equal and opposite reaction.

Mathematically,

$\vec{F}_1 = -\vec{F}_2$, where F_1 is the force exerted on body ② due to body ① and F_2 is the force exerted on body ① due to body ②.



According to conservation of linear momentum

$$\vec{P}_1 + \vec{P}_2 = \vec{P}_1 + d\vec{P}_1 + \vec{P}_2 + d\vec{P}_2$$

$$d\vec{P}_1 + d\vec{P}_2 = 0$$

$$\boxed{d\vec{P}_1 = -d\vec{P}_2}$$

$$\frac{d\vec{P}_1}{dt} = -\frac{d\vec{P}_2}{dt}$$

$$\boxed{\vec{F}_1 = -\vec{F}_2}$$

Newton's third law.

Action and reaction forces always act on two different bodies.

Examples:

- Swimming
- rowing a boat
- firing by a gun
- Book on a table
- Mass attached to a rope

Impulse
Impulse acts for a period
e.g. in driving

Impulse force time

SI

DF

★. Impulse dimension

It is in

Impulse.

Impulsive force: It is the force which acts on a body for a very short period of time.

E.g.; In hitting a ball with a bat, in driving a nail using hammer

Impulse is defined as the product of force applied and the small time for which the force is applied.

$$\vec{I} = \vec{F} \times t$$

SI unit of impulse

$$= \text{N} \cdot \text{sec}.$$

$$= \text{kg m sec}^{-2} \cdot \text{sec}$$

$$= \text{kg m sec}^{-1}$$

$$\begin{aligned} \text{DF of Impulse} &= M^1 L^1 T^{-2} \times T^1 \\ &= M^1 L^1 T^{-1} \end{aligned}$$

★. Impulse and linear momentum have same dimensional formula.

Impulse Momentum Theorem.

It states that the impulse applied to a body is equal to change in its linear momentum

$$\vec{I} = \vec{F} \times t$$

$$\vec{I} = m \cdot \vec{a} \times t$$

we have,

$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{t}$$

$$\vec{a} \times t = \vec{v}_f - \vec{v}_i$$

$$\therefore \vec{I} = m(\vec{v}_f - \vec{v}_i)$$

$$\vec{I} = m\vec{v}_f - m\vec{v}_i$$

$$\boxed{\vec{I} = \vec{p}_f - \vec{p}_i}$$

clearly, impulse gives us the effect of force on a body.

- Q. A ball of mass 0.5 kg is thrown towards a wall with a velocity of 10 m/s. The ball is reflected back by the wall with a velocity of 8 m/s. Find the impulse.

$$m = 0.5 \text{ kg}$$

$$\vec{v}_i = 10 \text{ m/s}$$

$$\vec{v}_f = -8 \text{ m/s}$$

$$\vec{p}_f = m\vec{v}_f = 0.5 \times 8 = -4 \text{ kgm/s}$$

$$\vec{p}_i = m\vec{v}_i = 5 \text{ kgm/s}$$

$$I = -4 - 5 = -9 \text{ kgm/s}$$

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Q A bullet of mass 200 g (0.2 kg) is fired from a gun with a mass of 20 kg with a velocity of 50 m/s. Find the velocity of recoil of gun.

$$\vec{v}_g = - \frac{m_b \vec{v}_b}{m_g}$$

$$\vec{v}_g = - \frac{10}{20} = -0.5 \text{ m/s.}$$

Q A man of mass 60 kg runs with a velocity of 2 m/s and suddenly jumps on a cart of mass 200 kg at rest. Find the common velocity with which the cart and the man will move.

mass of man = 60 kg.

velocity = 2 m/s.

$$\vec{p} = 120 \text{ kgm/s.}$$

Mass of cart = 200 kg.

After jumping mass = 260 kg.

$$\therefore \vec{v}_c = \frac{m_m \vec{v}_m}{m_c}$$

$$\vec{v}_c = \frac{120}{260} = 0.46$$

$$\begin{aligned} (60 \times 2) + (200 \times 0) &= 260 \text{ V} \\ 120 + 0 &= 260 \text{ V} \end{aligned}$$

$$V = \frac{120}{260}$$

$$V = 0.46 \text{ m/s}$$

Newton's second law of motion is known as real law of motion as both the first law and the third law are contained in it.

Apparent weight of a man in an elevator (lift).

NOTE: Weighing balance actually measures normal reaction ($R = mg$).

1. When the elevator is at rest or moving with constant velocity upwards or downwards, the ~~actual~~ apparent weight of the body is equal to its actual weight.

$$W = mg$$

2. When the elevator moves upwards with acceleration 'a' the apparent weight of the body is greater than its actual weight.

$$W = m(g + a)$$

3. When the elevator moves downwards with acceleration 'a' the apparent weight of the body is less than its

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$$w = m(g + (-a))$$

actual weight

$$w' = m(g - a)$$

If the elevator is in free fall

$$a = g$$

$$w = 0$$

Friction: It is the opposing force which comes into play when some force is applied on a body and the body either moves or tries to move over the surface of another body.

Types of friction:

1 **Static friction:** It is the friction which comes into play when some force is applied on a body and the body remains at rest.

2 **Limiting friction:** It is the friction which comes into play when some force is applied on a body and the body is at verge to move.

* Limiting friction represents maximum value of static friction.

3 **Kinetic friction:** It is the friction which comes into play when some force is applied on a body and the body actually moves over the surface of another body.

Old theory of friction.
According to this theory, it is the roughness of two surfaces in contact which is responsible for friction between them.

According to this theory, more the roughness of two surfaces in contact, more should be the friction and

→ smoother the two surfaces in contact, less should be the friction. but in case of two glass sheets in contact, although the surfaces are extra smooth but friction is more.

→ Also, acc to this theory, more the area of contact b/w two surfaces, more should be the friction. and less the area of contact less should be the friction. but it has been found that friction is independent of area of contact.

Modern Theory of Friction.

According to this theory it is the inter-molecular force of attraction b/w the two surfaces in contact which is responsible for friction between them.

Law

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3.

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____/____/____ Laws of limiting friction.

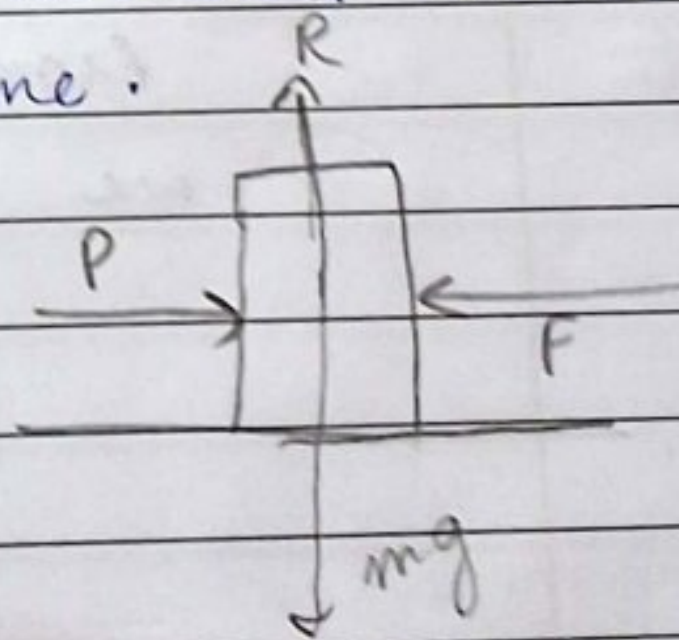
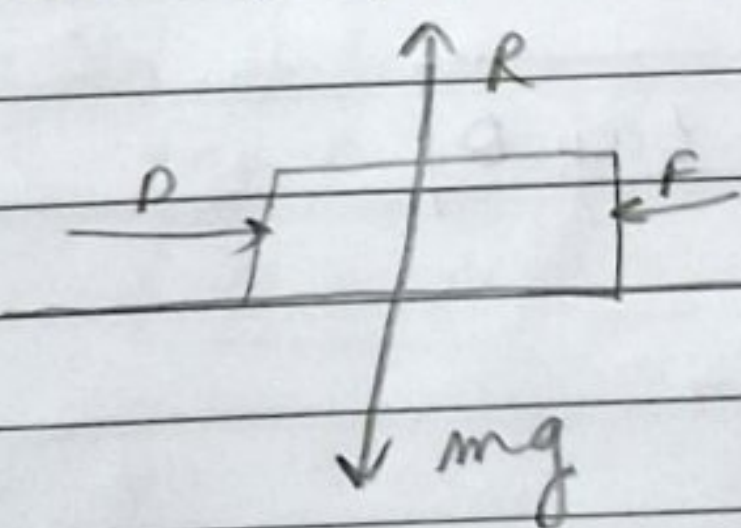
1. The force of limiting friction is directed opposite to the direction of applied force.
2. The force of limiting friction is directly proportional to normal reaction.

$$F \propto R$$

$F = \mu R$, where μ is a constant for the two surfaces in contact and is known as coefficient of limiting friction. Coefficient of limiting friction depends on the material of two surfaces in contact and the nature of the two surfaces in contact i.e.; whether the surfaces are polished or unpolished, dry or wet.

Coefficient of limiting friction is unitless as well as dimensionless.

3. The force of limiting friction is independent of apparent area of contact between two surfaces provided normal reaction (R) remains same.



4. The force of limiting friction depends on the nature and material of two surfaces in contact.

* Rolling friction is always less than sliding friction, that is why it is easy to roll a body instead of sliding it.

Imp. Angle of Friction (θ)
It is defined as the angle which the resultant of normal reaction (R) and force of limiting friction makes with normal reaction.

In $\triangle OCB$

$$\tan \theta = \frac{BC}{OC}$$

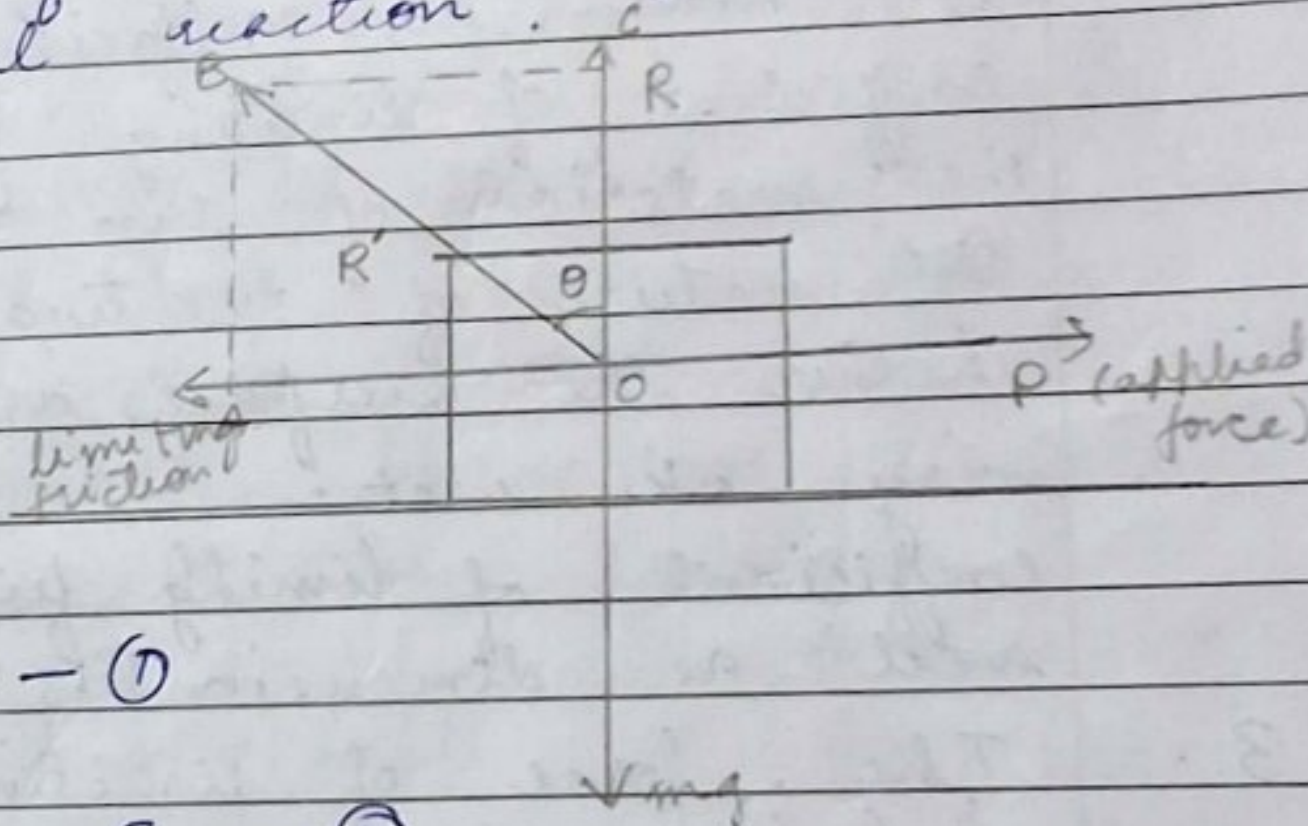
$$\tan \theta = \frac{OA}{OC}$$

$$\tan \theta = \frac{F}{R} \quad \text{--- (1)}$$

$$\text{Also } \mu = \frac{F}{R} \quad \text{--- (2)}$$

from eq (1) and (2)
we have.

$$\boxed{\mu = \tan \theta}$$



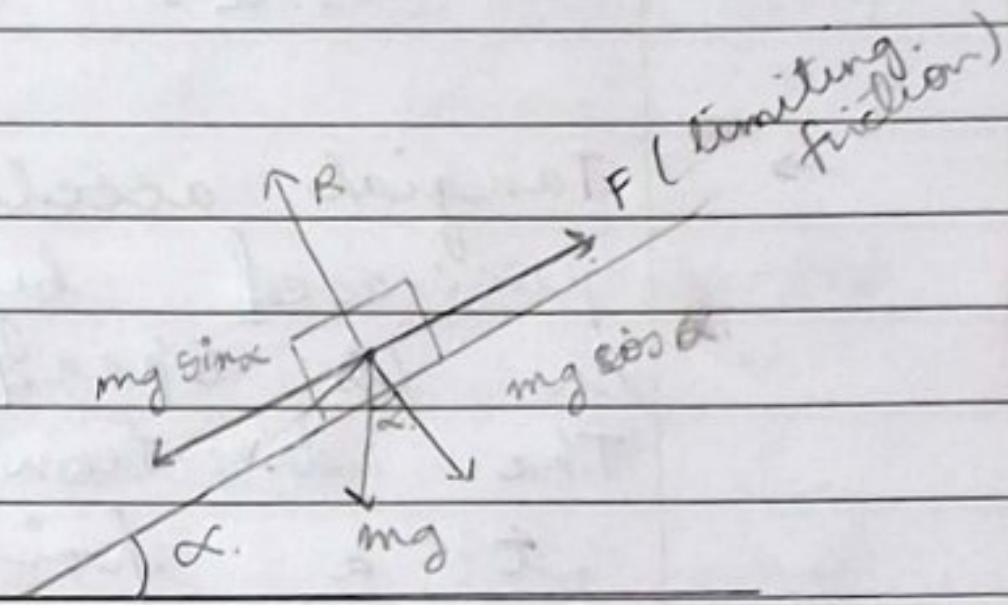
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Angle of Repose / Angle of sliding (α)
 It is defined as the angle of inclination at which a body placed on an inclined plane just begins to slide down. Above angle of repose, a body placed on an inclined plane slides down easily.

$$R = mg \cos \alpha$$

$$F = mg \sin \alpha$$

$$\frac{F}{R} = \frac{mg \sin \alpha}{mg \cos \alpha}$$



$$\text{Also, } \frac{F}{R} = \mu$$

$$\therefore \mu = \frac{\sin \alpha}{\cos \alpha}$$

$$\boxed{\mu = \tan \alpha}$$

$$\mu = \tan \theta$$

$$\text{Also } \mu = \tan \alpha$$

$$\tan \theta = \tan \alpha$$

$$\boxed{\theta = \alpha}$$

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circular motion.

uniform circular motion: circular motion in which speed of particle remains constant.

* In circular motion velocity is always variable.

→ Tangential acceleration: It is the acceleration possessed by a particle due to change in its speed (magnitude of its velocity). The direction of tangential acceleration at a point is given by a tangent drawn at that point.

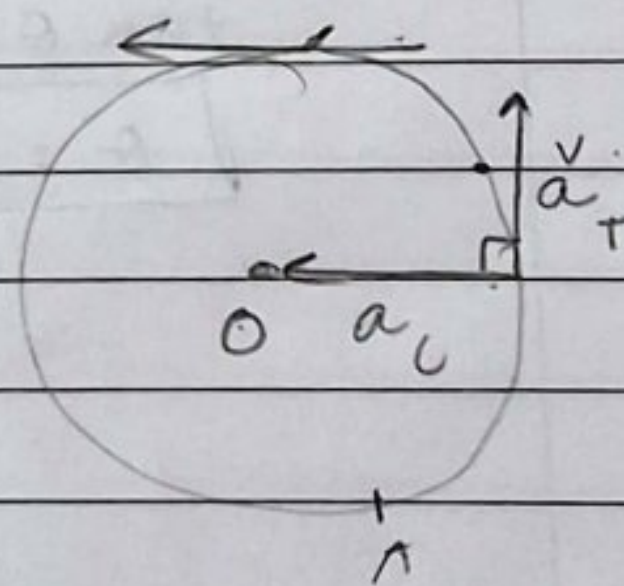
→ centripetal acceleration: It is the acceleration possessed by a particle due to the change in the direction of its velocity. Centripetal acceleration is directed towards the centre of the circle.

The angle b/w tangential acceleration and centripetal acceleration is 90° .

Net acceleration

$$a = \sqrt{a_t^2 + a_c^2} + 0$$

$$a = \sqrt{a_t^2 + a_c^2}$$



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The force responsible for moving a particle in a circle or curve is known as centripetal force and the acceleration possessed by a particle is known as centripetal acceleration.

The centripetal force is given by $F = \frac{mv^2}{r}$, where m is the mass of the particle, v is its velocity and r is the radius of the circle.

The centripetal force is always directed towards the centre of the circle. Comparing the above equation with $F = ma$, clearly $a = \frac{v^2}{r}$ and it is known as centripetal acceleration.

Vehicle on a level circular road.

Here frictional force acts as a centripetal force.

$$F = \frac{mv^2}{r} \quad \text{--- (1)}$$

Also $F = \mu R$.

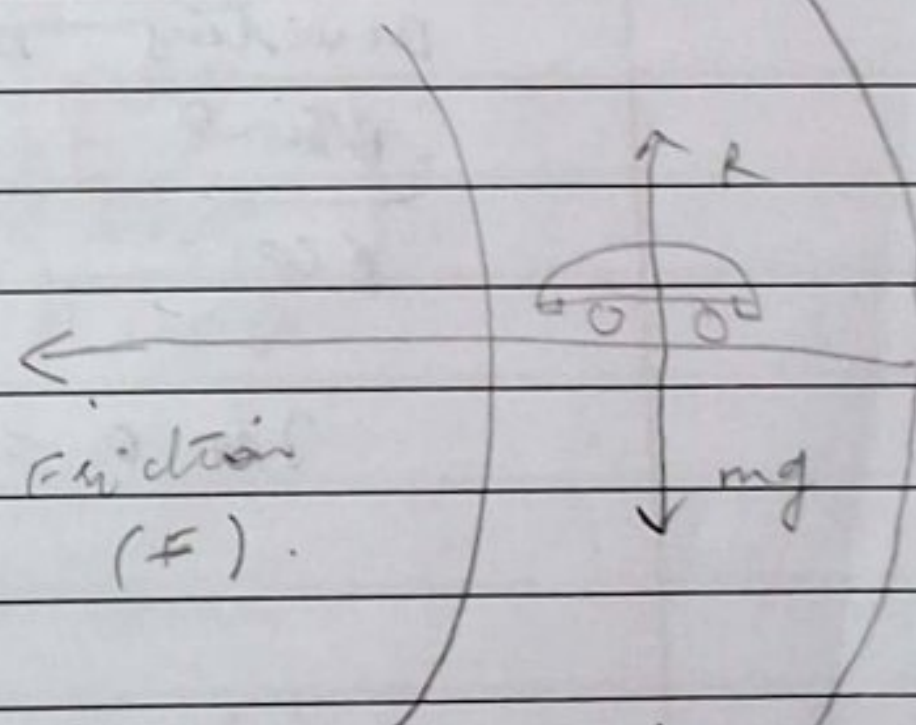
$$F = \mu mg \quad \text{--- (2)}$$

From (1) and (2)

$$\frac{mv^2}{r} = \mu mg$$

$$v^2 = \mu rg$$

$$v = \sqrt{\mu rg}$$



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This is the maximum safe velocity with which a vehicle can pass on a level circular road without skidding off from the road.

Banking of Roads (neglecting friction)
Raising the outer edge of a curved road is known as banking and the angle through which the edge is raised is known as angle of banking.

we have

$$R \cos \theta = mg \quad \text{--- (1)}$$

and $R \sin \theta$ acts as a centripetal force

$$\therefore R \sin \theta = \frac{mv^2}{r} \quad \text{--- (2)}$$

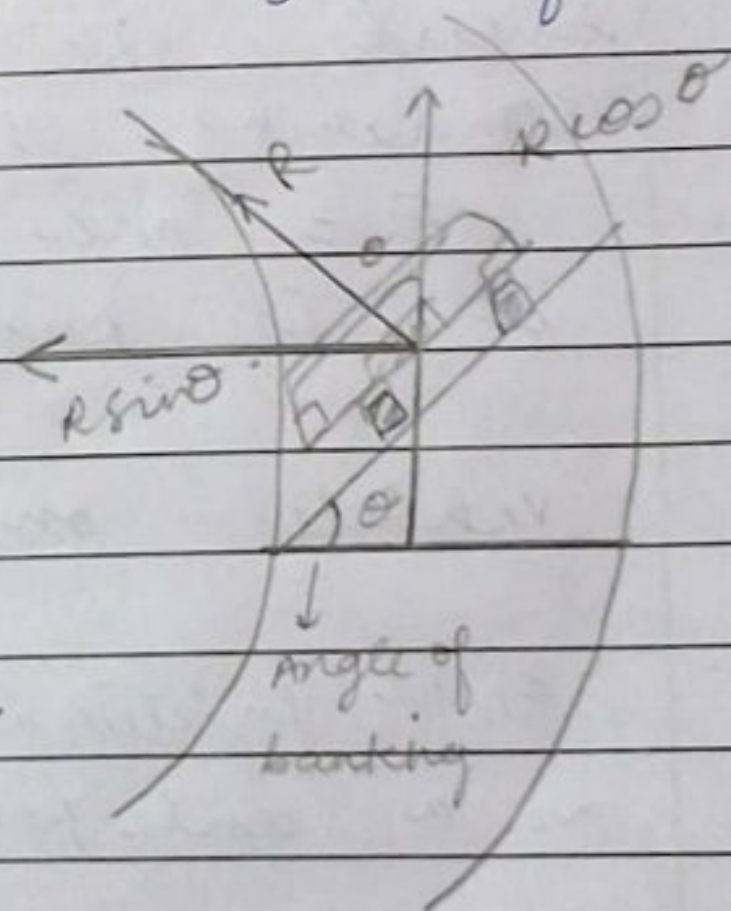
Dividing eq (2) by (1).

$$\frac{R \sin \theta}{R \cos \theta} = \frac{mv^2}{r} \times \frac{1}{mg}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$v^2 = rg \tan \theta$$

$v = \sqrt{rg \tan \theta}$ — This is the max velocity with which one can pass through a banked road safely.



This equation is also valid in case of bending of a cyclist. (θ = angle of bending)

Banking of roads (friction is considered).
we have,

$$mg + F \sin \theta = R \cos \theta.$$

$$mg = R \cos \theta - F \sin \theta \quad \text{--- (1)}$$

$$\text{Also } \frac{mv^2}{r} = R \sin \theta + F \cos \theta \quad \text{--- (2)}$$

Multiplying eqn (2) by eqn (1) F

$$\frac{v^2}{r} \cdot mg = \frac{R \sin \theta + F \cos \theta}{R \cos \theta - F \sin \theta}$$

$$\frac{v^2}{rg} = \frac{R \sin \theta + F \cos \theta}{R \cos \theta - F \sin \theta}$$

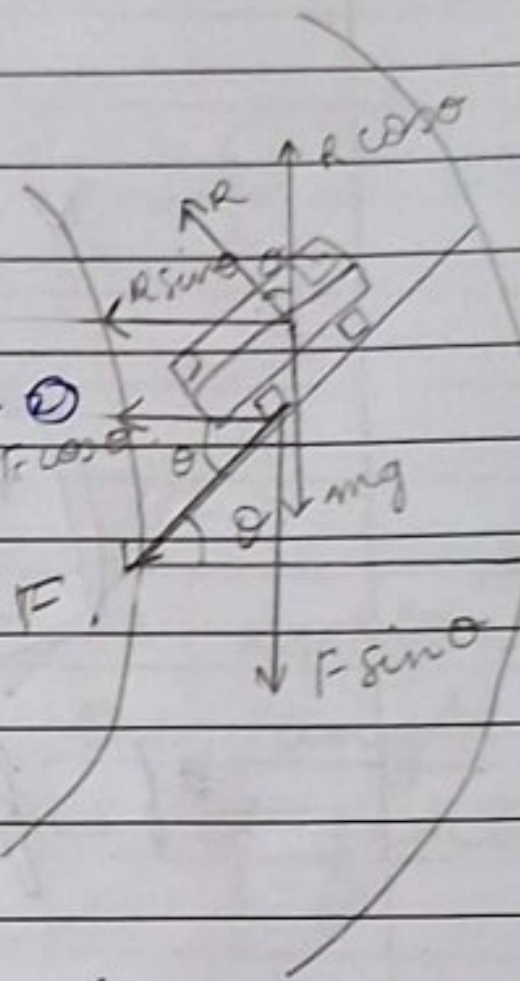
$$\frac{v^2}{rg} = \frac{\frac{R \sin \theta}{R \cos \theta} + \frac{F \cos \theta}{R \cos \theta}}{\frac{R \cos \theta}{R \cos \theta} - \frac{F \sin \theta}{R \cos \theta}}$$

$$\frac{v^2}{rg} = \frac{\tan \theta + \mu}{1 - \mu \tan \theta}$$

$$v^2 = rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$$

$$v = \sqrt{rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)}$$

$$v = \left[rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right) \right]^{\frac{1}{2}}$$



* For a level curved road.
 $\theta = 0$

$$v = \sqrt{rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)}$$

$$v = \sqrt{rg \left(\frac{0 + \mu}{1 - 0} \right)}$$

$$v = \sqrt{rg \mu} = \sqrt{rg}$$

* If friction is neglected,
 $\mu = 0$

$$v = \sqrt{rg \left(\frac{\tan \theta + 0}{1 - 0(\tan \theta)} \right)}$$

$$v = \sqrt{rg \tan \theta}$$

Q. A man of mass 60 kg is standing in a lift which is moving downwards with an acceleration of 9.8 m/s^2 . The apparent weight of the man is

- a) 60 N
- b) 0 N
- c) 100 N
- d) 110 N

$$W = m(g - a)$$

$$W = 60(9.8 - 9.8)$$

$$W = 0 \text{ N}$$

Q.

A body of mass 2kg starting from rest attains a velocity of 6 m/s in 2 sec. Find the force on the body.

$$F = m \cdot a$$

$$a = \frac{v - u}{t} = \frac{6 - 0}{2} = \frac{6}{2} = 3\text{ m/s}^2$$

$$F = 2 \times 3 = 6\text{ N}.$$

Q.

The angle of friction for 2 surfaces in contact is 60° . Find the angle of repose and the coefficient of friction (μ).

$$\theta = 60^\circ.$$

Angle of repose.

$$\theta = \alpha = 60^\circ.$$

Also coefficient of friction.

$$\mu = \tan 60^\circ.$$

$$\mu = \sqrt{3} = 1.73.$$

Q.

A force of 10N is applied on a body of mass 4kg and the body is about to start moving. Find the coefficient of friction. (take $g = 10\text{ m/s}^2$).

$$mg = R = 40\text{ N}.$$

$$F = 10\text{ N}.$$

$$\therefore \mu = \frac{F}{R} = \frac{10}{40} = 0.25.$$